

2018-Ph-H1-Q20

Section: Our Dynamic Universe

Topic: Collisions, Explosions and Impulse

Two trolleys collide and stick together.

Before the collision, one has momentum of $6 \text{ kg}\cdot\text{m/s}$, and the other $2 \text{ kg}\cdot\text{m/s}$ in the opposite direction.

What is their combined momentum after the collision?

Solution:

Total momentum:

$$6 + (-2) = 4 \text{ kgm/s}$$

(Momentum is a vector — direction matters.)

Answer: $4 \text{ kg}\cdot\text{m/s}$

Revision Tip:

Add momenta **with signs** — assign one direction as positive. Total momentum is conserved in collisions.